

Introduction: Open-Vocabulary Object Detection

Traditional object detection (closed-set)

- Detects only a fixed set of categories seen during training
- Example: {person, car, dog, ...}

Open-vocabulary object detection

- Uses text queries to detect categories beyond the fixed training label set
- Example queries: *"computer case handle"*, *"CMOS battery"*, *"sticker on top of the computer case"*

Key challenge

- Difficult to detect rare objects and fine-grained parts, especially when they are small.

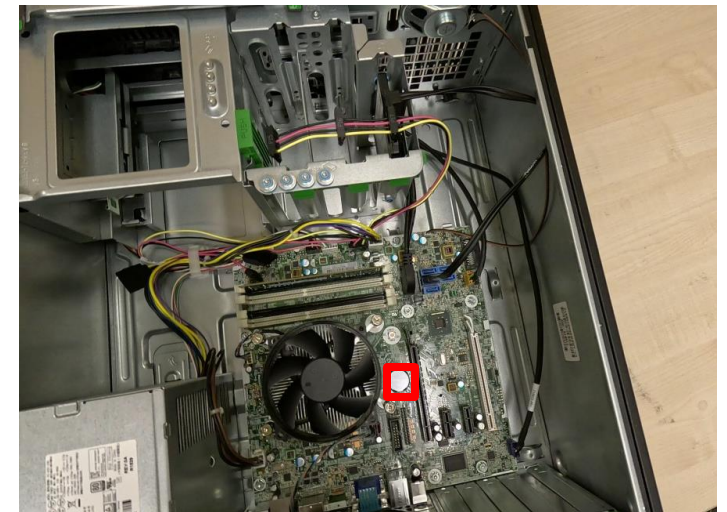


Figure 1. The aim is to detect rare object parts and components such as these.

Related work

Methods

1. *YOLO-World: Real-Time Open-Vocabulary Object Detection*, Tianheng Cheng et al., CVPR 2024 — [[paper](#), [code](#)]
2. *LLMDet: Learning Strong Open-Vocabulary Object Detectors under the Supervision of Large Language Models*, Shenghao Fu et al., CVPR 2025 (Highlight) — [[paper](#), [code](#)]
3. *DINO-X: A Unified Vision Model for Open-World Object Detection and Understanding*, Tianhe Ren et al., arXiv 2024 — [[paper](#), [code](#)]

Datasets

1. *LVIS: A Dataset for Large Vocabulary Instance Segmentation*, Agrim Gupta et al., CVPR 2019 — [[paper](#), [dataset](#)]
2. *PACO: Parts and Attributes of Common Objects*, Vignesh Ramanathan et al., CVPR 2023 — [[paper](#), [dataset](#)]

Related work

Strengths and weaknesses

- Existing methods (e.g., YOLO-World, LLMDet, DINO-X) work well for generic objects and phrases, such as “a man wearing a blue t-shirt”.
- However, they struggle at detecting rare, small, and fine-grained object parts.

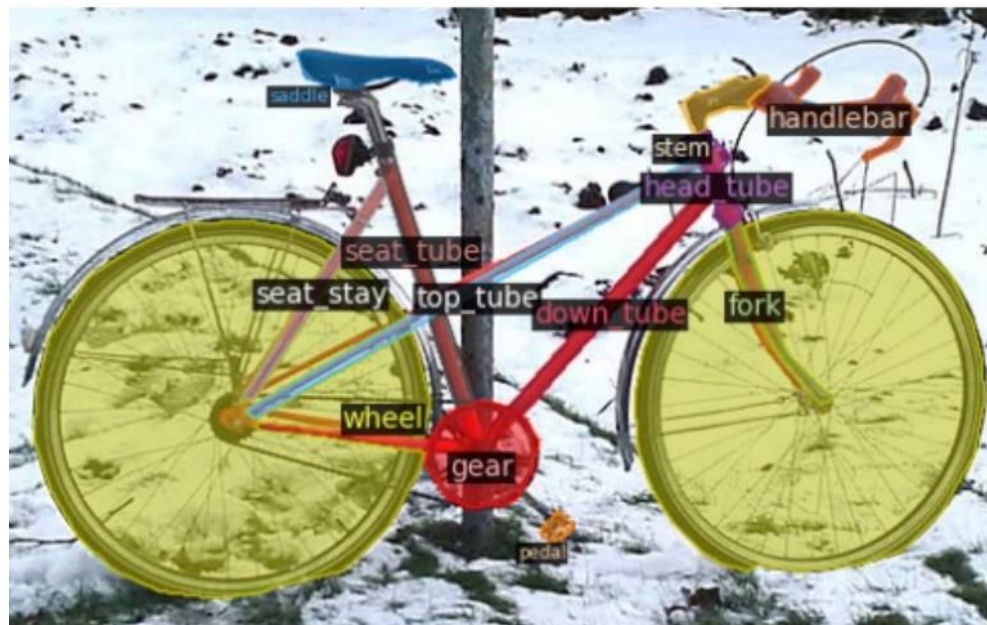
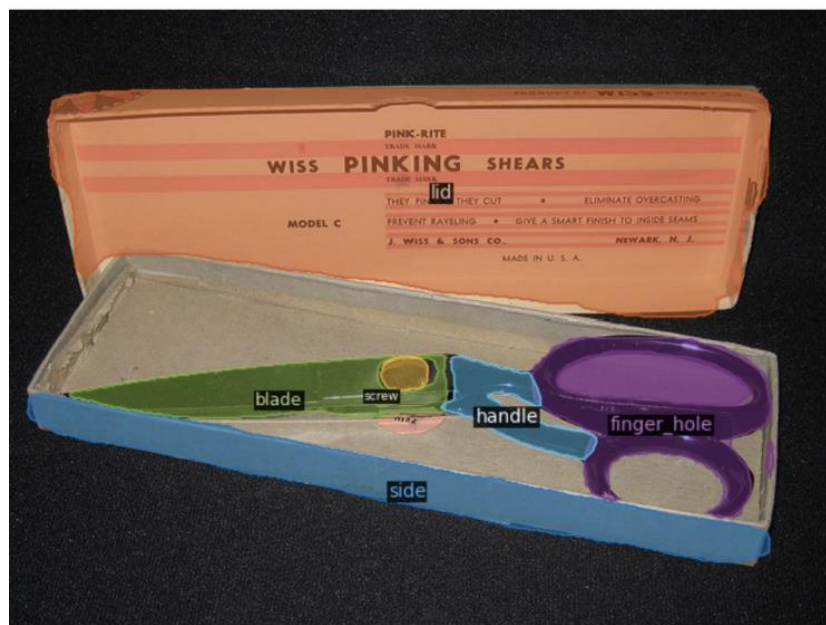


Figure 2. Annotation examples from the PACO dataset.

Project work

1. Data preparation
 - Use standard datasets, e.g., PACO-EGO4D
 - Collect and annotate a small custom dataset
2. Run baseline approaches
 - YOLO-World, DINO-X, etc. (1-2 approaches)
 - Test commercial VLMs, e.g., ChatGPT or Gemini
 - Optionally test open-source VLMs
3. Implement an agentic VLM approach
 - See the next slide for details
4. Compare against baselines
 - Evaluate on both standard and custom datasets
 - Analyze strengths, weaknesses, and failure cases

Agentic VLM approach

The idea is to use a VLM in an iterative loop: **propose** → **verify** → **refine**

1. Initial query

- Pass the full image and the text query to the VLM
- Obtain a rough candidate region for the target part

2. Crop-based refinement

- Crop the candidate region from the original image
- Re-run the VLM on the crop to focus on small details

3. Bounding box verification

- Overlay the candidate bounding box on the image or crop
- Ask the VLM whether the box covers the queried part
- The system should also recognize if the queried part is absent

4. Iterative update

- If the box is not correct, propose a refined box
- Repeat until the box is accepted or rejected

Note: A coordinate grid overlay could also be explored to improve spatial reasoning

Agentic VLM approach

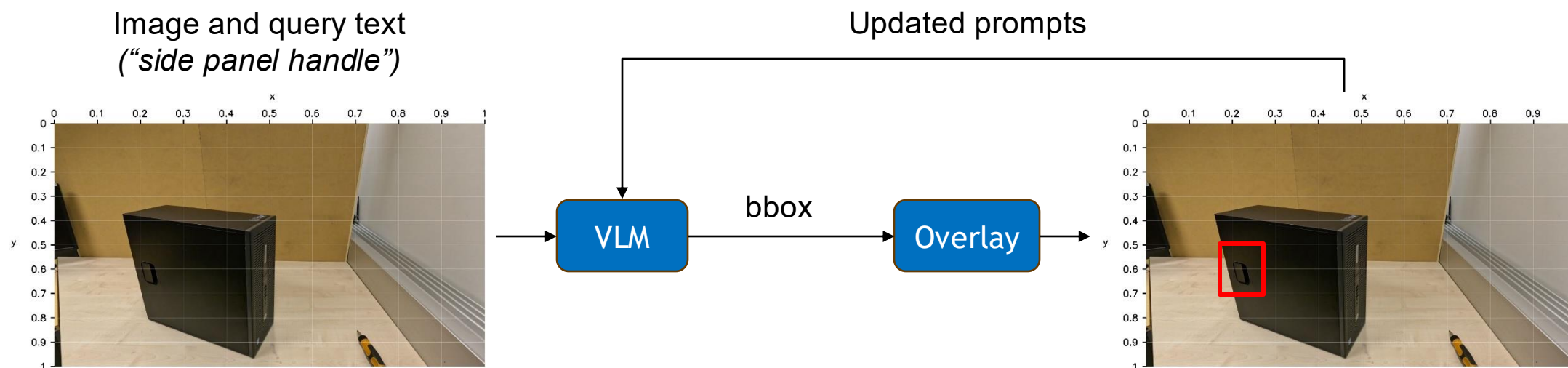


Figure 3. Agentic VLM pipeline: the image is augmented with a coordinate grid, the VLM predicts a candidate bounding box, and the prediction is iteratively verified and refined using overlay-based prompts.